# Supplementary Material for Semi-discrete Optimal Transport for Time-Varying Multi-Agent Coverage Control

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## 1 Simulation details

Videos of the simulations can be found at the Github repository https://shorturl.at/vgPu8.

We first list the experiments. Experiments 1 and 2 are presented in Sec. 5 of the manuscript, while Experiments 3–6 are included in Sec. 3 of this supplementary material due to space constraints in the manuscript.

**Experiment 1** is the 2D experiment presented in Sec. 5, and its results are shown in Figures 1–2 of the manuscript.

**Experiment 2** is the 1D experiment presented in Sec. 5, and its results are shown in Figure 3 of the manuscript.

**Experiment 3** is an additional simulation, shown below, where 5 agents track a time-varying *bimodal* Gaussian distribution. Its discussion and results are presented in Section 3 of this document.

**Experiment 4** consists of an additional set of simulations assessing the benefits of incorporating the feedforward term for different values of the gain $K_x$ and the reference speed $\dot{m}$, corresponding to the velocity of the mean of the Gaussian reference distribution. Its discussions and results are presented in Section 3 of this document.

**Experiment 5** evaluates the robustness of the proposed method to uncertainties in $\partial_t \bar{\rho}$ for different values of $K_x$. Specifically, uncertainty is introduced by perturbing the velocity $\dot{m}$ of the Gaussian reference mean by $\pm 20\%$. Its discussions and results are presented in Section 3 of this document.

**Experiment 6** investigates the scaling properties of the proposed method with respect to the number of agents $N$. The results show that the 2-Wasserstein distance in one dimension decreases proportionally to $1/N$, consistently with optimal quantization theory [4]. Its discussions and results are presented in Section 3 of this document.

All simulations were performed using MATLAB. Integrations were performed via the standard forward Euler method, truncated after some time $T$.

Table 1 contains the parameters for the 2D experiment. Here, the time-varying reference is initially Gaussian and splits into an equally weighted mixture of two Gaussians at $t = \pi$:

$$\bar{\rho} = \begin{cases} \mathcal{N}(\boldsymbol{\mu}(t), \boldsymbol{\Sigma}(t)), & 0 \le t < \pi, \\ \frac{1}{2}\mathcal{N}(\boldsymbol{\mu}_1(t), \boldsymbol{\Sigma}(t)) + \frac{1}{2}\mathcal{N}(\boldsymbol{\mu}_2(t), \boldsymbol{\Sigma}(t)), & t \ge \pi, \end{cases}$$

where $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes the Gaussian density with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. The mean trajectories are

$$\boldsymbol{\mu}(t) = \begin{bmatrix} 10 + 6t \\ 10 + 6t \end{bmatrix}, \qquad \boldsymbol{\mu}_1(t) = \begin{bmatrix} 10 + 6\pi - 6(t - \pi) \\ 10 + 6\pi \end{bmatrix},$$

$$\boldsymbol{\mu}_2(t) = \begin{bmatrix} 10 + 6\pi \\ 10 + 6\pi - 6(t - \pi) \end{bmatrix},$$

and the common covariance is

$$\boldsymbol{\Sigma}(t) = \begin{cases} \big(15 + 10\sin(2t)\big)\mathbf{I}_2, & 0 \le t < \pi, \\ 15 - 10(t - \pi)\mathbf{I}_2, & t \ge \pi. \end{cases}$$

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Thus, the initial mode moves diagonally before splitting into two modes moving horizontally and vertically, respectively, while their variances decrease after the split.

Table 1
Parameters used in Experiment 1.

| Parameter | Value | Parameter | Value |
|---|---|---|---|
| $\Omega$ | $[0, 50]^2$ | $\gamma_1$ | 9 |
| $\Delta t$ | 0.01 | $\gamma_2$ | 10 |
| $\Delta x$ | 0.5 | $\boldsymbol{m}_0$ | $[10, 10]$ |
| $K_x$ | 10 | $\boldsymbol{\Sigma}_0$ | $\mathrm{diag}(15, 15)$ |
| $K_\phi$ | 10 | $\boldsymbol{z}_1(0)$ | $\mathbf{0}_{2N}$ |
| $\boldsymbol{z}_0(0)$ | $\mathbf{0}_{2N}$ | | |

The feedforward term $\widehat{\dot{\mathbf{b}}}_i$ in Experiment 1 is estimated via the second-order sliding mode differentiator found in [3] in order to cope with noisy signals:

$$\begin{cases} \dot{\mathbf{z}}_0 = -\gamma_1 |\mathbf{z}_0 - \mathbf{b}|^{1/2} \mathrm{sign}(\mathbf{z}_0 - \mathbf{b}) + \mathbf{z}_1, \\ \dot{\mathbf{z}}_1 = -\gamma_2 \, \mathrm{sign}(\mathbf{z}_0 - \mathbf{b}), \end{cases} \tag{1}$$

yielding $\widehat{\dot{\mathbf{b}}}(t) = \mathbf{z}_1(t)$. We remark that this numerical differentiator approximates $\dot{\mathbf{b}}_i(t)$ in a distributed fashion. In TVOT-C, the condition number of the matrix $\mathbf{M}$ reaches its maximum value of approximately $10^9$ at the final simulated time. As the two modes of the Gaussian reference separate, the edges connecting cells around different modes pass through a region of very low density, weakening their coupling. Nonetheless, the matrix remains numerically invertible along the simulated trajectory, and the simulation completes without solver errors: although the condition number is high, it remains well below the reciprocal of the double-precision machine epsilon ($\approx 4.5 \times 10^{15}$). In the distributed controllers, the spectral radius of $\widehat{\mathbf{M}}$ approaches 1 towards the final time. Since the controller uses a fixed, finite truncation, convergence of the infinite Neumann series is not guaranteed; nevertheless, the chosen truncation yields satisfactory tracking in the reported experiment.

Figure 1 shows the Wasserstein error over time of the proposed solutions related to Experiment 1. As quantified in Sec. 2, decentralized approximations yield comparable results to the centralized one, which varies sinusoidally until $t = \pi$, because the variance of the Gaussian reference varies sinusoidally as well, and after a short transient reaches a plateau (for $t > \pi$ the variance is constant).

Figures 2, 3, 3, 6, and 5 show snapshots of the TVOT-C, TVOT-$D_1$, TVOT-$D_4$, TVV-C, and OTCC solutions, respectively.

Table 2 contains the parameters for the 1D experiments.

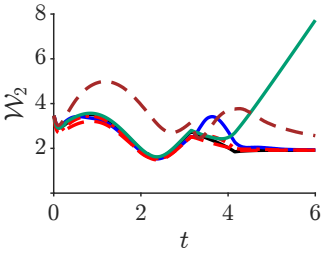



Fig. 1. Wasserstein errors over time for the 2D representative example using TVOT-C (black solid lines); TVOT-$D_4$ (dashed red line); TVOT-$D_1$ (dash-dotted red line); and TVOT-G (solid blue line); TVV-C (green solid line); OTCC (brown dashed line).

Table 2
Parameters used in the 1D experiments.

| Parameter | Exp. 2 | Exp. 3 |
|---|---|---|
| $\Omega$ | $[0, 50]$ | $[0, 50]$ |
| $K_x$ | 10 | 1 |
| $K_\phi$ | 10 | 1 |
| $\Delta t$ | 0.01 | 0.01 |
| $\Delta x$ | 0.01 | 0.01 |
| $m_0$ | 20 | $(10, 20)$ |
| $\sigma_0$ | 3 | $(2, 2)$ |
| $\dot{m}$ | $-5\sin(t)$ | $(1, 1)$ |
| $\dot{\sigma}$ | $\sin(t)$ | $(0, 0)$ |

The feedforward term $\widehat{\dot{\mathbf{b}}}_i$ in Experiment 2 is estimated via finite differences

$$\widehat{\dot{b}}_i(t) = \frac{b_i(t) - b_i(t - \Delta t)}{\Delta t}.$$

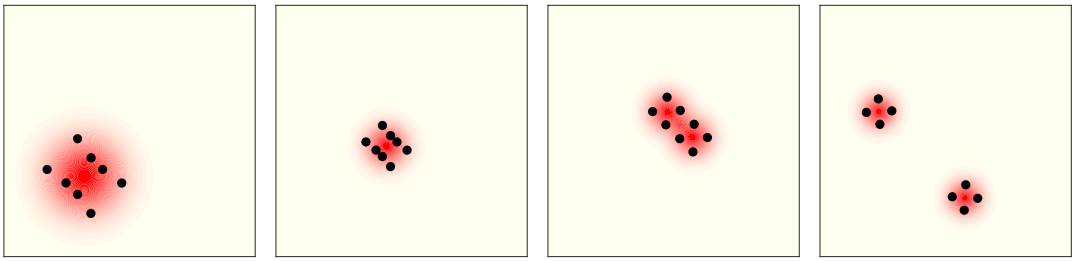
In Experiment 2, the maximum condition number of $\mathbf{M}$ is 55 for the TVOT-C solution, indicating no issues in inverting it. In the TVOT-$D_1$ simulation, the spectral radius of $\widehat{\mathbf{M}}$ is always strictly lower than 1, satisfying the underlying convergence assumption.

## 2 Quantitative Comparison

Next, we provide a quantitative comparison of the experiments reported in the manuscript to substantiate the claims made in Section 5 of the manuscript (Table 3). Discussion of the main findings is present in Sec. 5 of the manuscript.

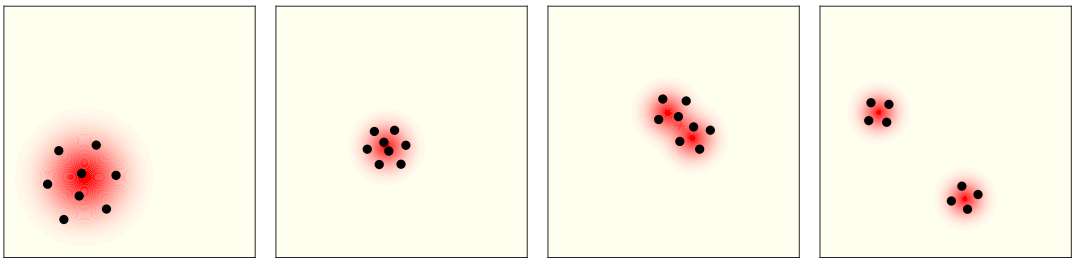
## 3 Additional Simulations

Experiment 3 represents a validation in a one-dimensional domain, where 5 agents track a bimodal Gaussian



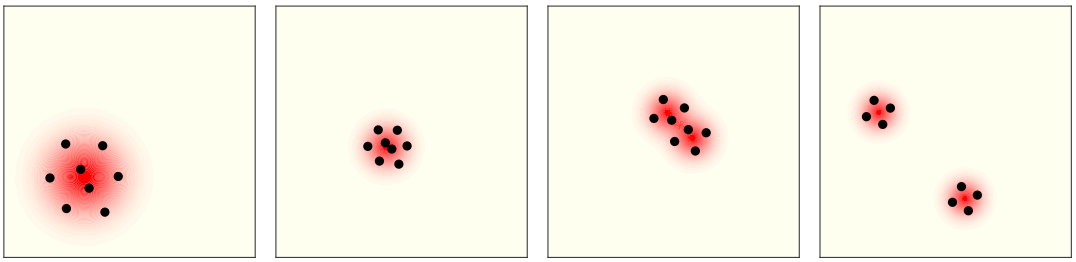
(a) (b) (c) (d)

Fig. 2. Validation of the control law TVOT-C with 8 agents (black dots) at (a) $t = 1$, (b) $t = 2$, (c) $t = 4$, and (d) $t = 6$ a.u.



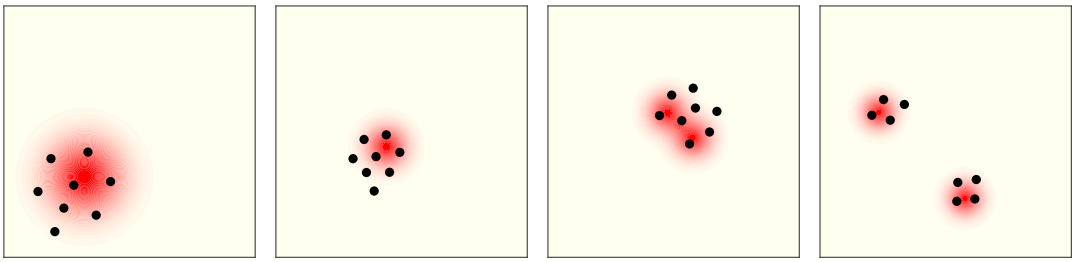
(a) (b) (c) (d)

Fig. 3. Validation of the control law TVOT-$D_1$ with 8 agents (black dots) at (a) $t = 1$, (b) $t = 2$, (c) $t = 4$, and (d) $t = 6$ a.u.



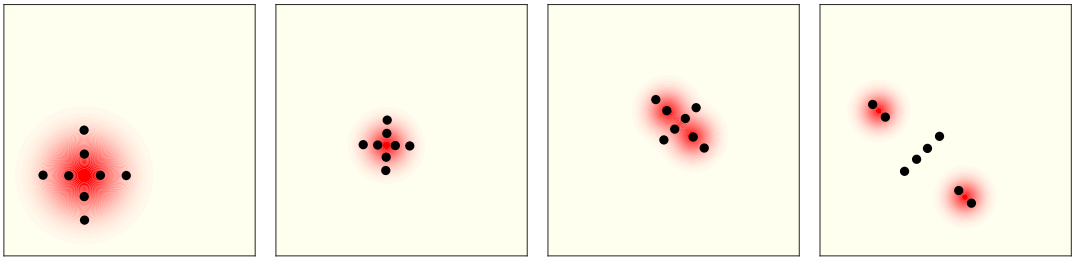
(a) (b) (c) (d)

Fig. 4. Validation of the control law TVOT-$D_4$ with 8 agents (black dots) at (a) $t = 1$, (b) $t = 2$, (c) $t = 4$, and (d) $t = 6$ a.u.



(a) (b) (c) (d)

Fig. 5. Validation of the control law OTCC with 8 agents (black dots) at (a) $t = 1$, (b) $t = 2$, (c) $t = 4$, and (d) $t = 6$ a.u.



(a) (b) (c) (d)

Fig. 6. Validation of the control law TVV-C with 8 agents (black dots) at (a) $t = 1$, (b) $t = 2$, (c) $t = 4$, and (d) $t = 6$ a.u.

Table 3
Numerical comparison of tracking errors and Wasserstein distance.

| Method | $\limsup_t \|\mathbf{e}_a\|$ | $\limsup_t \|\mathbf{e}_p\|$ | $\limsup_t W_2(\bar{\mu}_t, \mu_t)$ |
|---|---|---|---|
| *Experiment 1* | | | |
| TVOT-C | $6.2 \times 10^{-7}$ | $5.4 \times 10^{-4}$ | 1.91 |
| TVOT-G | $9.6 \times 10^{-2}$ | 1.6 | 1.93 |
| TVOT-$D_1$ | $3.9 \times 10^{-2}$ | $8.6 \times 10^{-1}$ | 1.92 |
| TVOT-$D_4$ | $2.7 \times 10^{-2}$ | $2.9 \times 10^{-1}$ | 1.92 |
| TVV-C | $3.5 \times 10^{-1}$ | $1.1 \times 10^{-4}$ | 7.73 |
| OTCC | $1.7 \times 10^{-1}$ | 4.1 | 2.56 |
| *Experiment 2* | | | |
| TVOT-C | $5.37 \times 10^{-4}$ | $9.38 \times 10^{-4}$ | 1.25 |
| TVOT-G | $8.36 \times 10^{-2}$ | $2.07 \times 10^{-2}$ | 1.28 |
| TVOT-$D_1$ | $2.05 \times 10^{-1}$ | $5.76 \times 10^{-1}$ | 1.60 |
| TVV-C | $1.76 \times 10^{-1}$ | $9.20 \times 10^{-3}$ | 1.60 |
| OTCC | $5.11 \times 10^{-1}$ | 1.8 | 5.17 |

distribution $0.5\mathcal{N}(m_1(t), \sigma) + 0.5\mathcal{N}(m_2(t), \sigma)$ with $\dot{m}_i(t) = v_{m,i}$. The agents split into two groups near the peaks, accurately reproducing the target density. In the centralized solution, the middle agent positions itself in the valley between the two modes to guarantee equal mass partition of the domain (Fig. 7a). All other solutions have uneven partitioning of the agents among the two modes (Fig. 7b–e), leading to much higher Wasserstein and residual errors (Fig. 7f–h). This major difference is consistent with the global result of Theorem 2.

The benefit of feedforward compensation is further quantified in Experiment 4, represented in Fig. 8, which compares TVOT-G and OTCC for varying density drift velocity $\dot{m} \in [0.5, 4]$ and gain $K_x \in [0.5, 4]$ of a target Gaussian distribution. Without feedforward, acceptable accuracy requires large gains and slow drift; otherwise, persistent lag produces substantial steady-state errors. With the feedforward term, TVOT-G achieves significantly smaller errors across the entire parameter range. Here, after the transient, we observe that the disturbance in the mass dynamics vanishes and TVOT-G achieves asymptotically zero mass error, approaching the same moving stationary configuration as TVOT-C up to a negligible approximation error.

Under the same setup of the previous experiment, robustness to uncertainty in $\partial_t \bar{\rho}$ is evaluated in Experiment 5, represented in Fig. 9a, where $\bar{\rho} = \mathcal{N}(m(t), \sigma^2)$, where perturbations of $\pm 20\%$ are introduced in the estimated velocity of the target density mean $\dot{m} = 5$. Increasing $K_x$ reduces sensitivity to estimation errors, and tracking error remains bounded for a suitable range of parameters.

Finally, scaling is examined in Experiment 6 by varying $N \in [50, 500]$. Fig. 9b shows that the 2-Wasserstein error in 1D decays approximately as $1/N$ (slope in log-log scale is $-0.99$), consistent with optimal quantization theory [4].

## References


[1] Daisuke Inoue, Yuji Ito, and Hiroaki Yoshida. Optimal transport-based coverage control for swarm robot systems: Generalization of the Voronoi tessellation-based method. *IEEE Control Systems Letters*, 5(4):1483–1488, 2021.

[2] Sung G. Lee, Yancy Diaz-Mercado, and Magnus Egerstedt. Multirobot control using time-varying density functions. *IEEE Transactions on Robotics*, 31(2):489–493, 2015.

[3] Arie Levant. Higher-order sliding modes, differentiation and output-feedback control. *International Journal of Control*, 76(9–10):924–941, 2003.

[4] Paul Zador. Asymptotic quantization error of continuous signals and the quantization dimension. *IEEE Transactions on Information Theory*, 28(2):139–149, 1982.

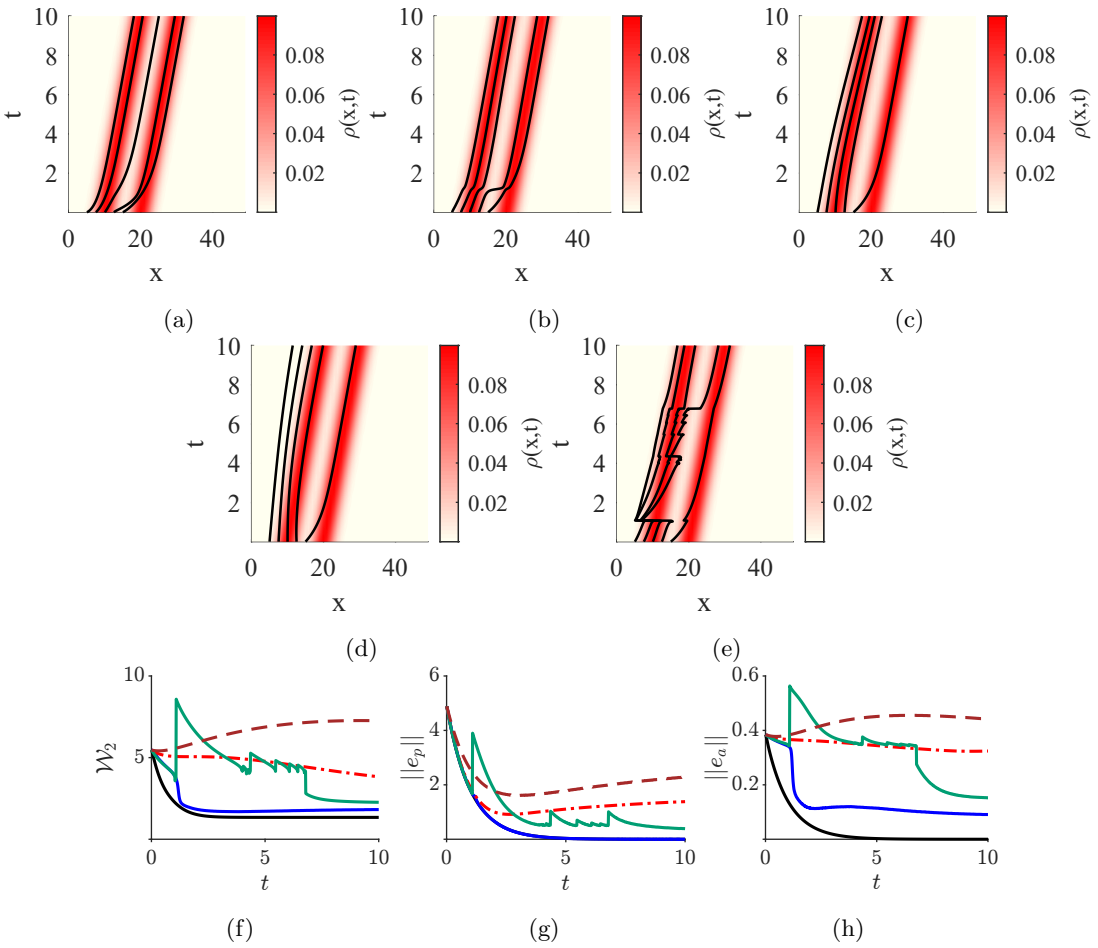



Fig. 7. Validation in one dimension with 5 agents (Experiment 3). Density evolution (background color from white to red) and agent trajectories (solid black lines) using (a) TVOT-C; (b) TVOT-G; (c) TVOT-D$_1$; (d) OTCC; and (e) TVV-C for the linearly varying mean *bimodal* Gaussian reference density. (f) Wasserstein error $\mathcal{W}_2$; (g) position error norm $\|\mathbf{e}_p\|$; and (h) mass error norm $\|\mathbf{e}_a\|$, for TVOT-C (black solid lines); TVOT-D$_1$ (dash-dotted red lines); TVOT-G (solid blue lines); OTCC [1] (dashed brown lines); TVV-C [2] (solid green lines).

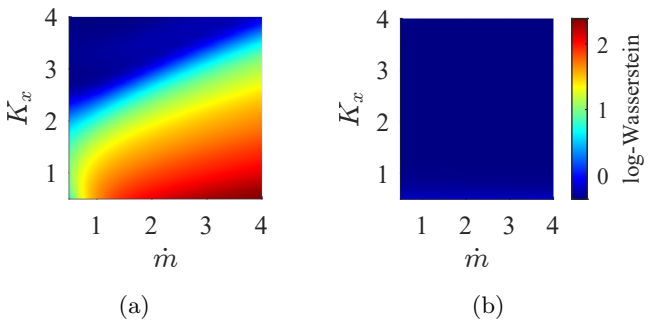



Fig. 8. Wasserstein error (log scale) as a function of density drift velocity (x-axis) and gain $K_x$ (y-axis) for (a) OTCC without feedforward and (b) the proposed TVOT-G strategy.

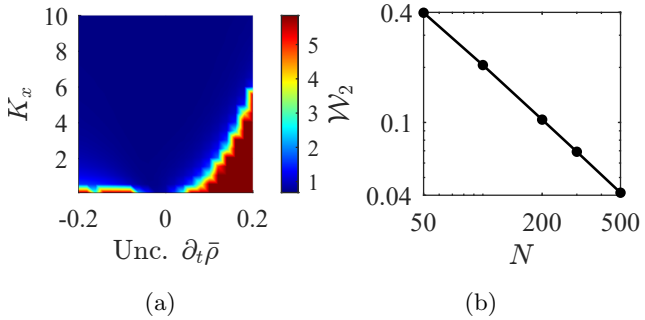



Fig. 9. (a) Robustness analysis showing Wasserstein error for varying estimation errors in $\partial_t\bar{\rho}(\mathbf{x},t)$ (x-axis) and gain $K_x$ (y-axis); the target mean velocity is 5 a.u. (b) Scaling analysis showing the 2-Wasserstein error at final time versus number of agents $N$ (log-log scale); the slope of approximately $-0.99$ indicates $\mathcal{W}_2 \sim 1/N$.